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(9630) PH04 Energy and energy resources
Report on the examination

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REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH04 Energy and energy resources JUNE 2023

This paper was similar to those of previous years in terms of demand and specification coverage. The questions gave students many opportunities to demonstrate their knowledge, understanding and skills across a range of topics. A variety of question styles including short answers, single and multi-step calculations, extended writing, and multiple choice were used.

The balance of assessment objectives and the mathematical demand was similar to previous series.

QUESTION 01

Most students were able to get one mark for this question. This was usually for stating that the molecules move randomly. Answers that linked the property to what was observed were very rare, so few students were given more than two marks. Many students took no account of the context of the question and simply listed the properties of molecules in an ideal gas. It was also often not clear whether a student was referring to the smoke particles or the gas molecules in their answers.

QUESTION 02

Few students had any difficulty achieving full marks for Q02.1. The design of the mark scheme helped to ensure that the question discriminated very well. Partial credit was available for incorrect answers that were sufficiently clear for examiners to see what the student was doing. The most common errors were linked to the use of the incorrect equation for the volume of a cylinder. Some problems converting the temperature to kelvin were also seen.

Question Q02.2 was only slightly more demanding and, again, the discrimination was high. The most common error was a failure to subtract the initial volume.

Most students were able to score one mark in Q02.3 for using $pV = nRT$, and referring to V and R as constants. Very few went any further than this. The best answers used data from **Figure 3** to show that the intercept of the line was $(0,0)$. Without this, referencing a straight line was insufficient to show that the temperature was constant.

In question Q02.4, many students attempted to use the first law of thermodynamics without realising that this was not an isolated system. The best answers made a clear statement about what was meant by the internal energy of an ideal gas and set out a logical sequence to explain why the internal energy must increase, in a very similar way to that given in the mark scheme. Students were expected to state that the potential energy of the molecules of gas was zero in order to gain full marks.

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QUESTION 03

Question 03.1 was one of the most discriminating questions on the paper. Some students showed their serious lack of understanding of this area by using the Boltzmann Constant for k in the conductivity equation. Several students were not given the alternative unit mark because there was no difference between their lowercase k in kilogram and their upper case K in kelvin.

Most of the students who obtained the first mark in Q03.2 went on to get the second mark too. The most common exceptions were students who rounded their answer incorrectly.

Question Q03.3 required students to consider the context of the question. Very many students who were not given the mark simply stated that a carpet would act as an insulator and therefore reduce the wasted heat transfer. The best answers included the idea that, as the running temperature was much higher when the carpet was in place, there would be a greater rate of heat transfer to the surroundings.

QUESTION 04

Most students were able to complete the calculation in Q04.1 correctly. Those who set out their answers clearly were able to pick up intermediate marks when they made an error. Examiners were looking for the important steps in the calculation to award credit. The discrimination in this question was the highest on the paper.

Many of the answers to question Q04.2 were vague or incomplete so very few students were able to gain the one mark available.

Question 04.3 also discriminated well, with almost one-third of students getting full marks. The best answers made a clear statement that, as the change in electrostatic potential energy (EPE) was the same, the nucleus with the greater charge (Y) would stop the alpha particle at a greater distance and therefore have a greater estimated radius. Partial answers often made no reference to the EPE or simply referred to the number of protons rather than the charge.

The curve in Q04.4 has been tested many times in the past but, often due to a lack of care when drawing, only about 40% of students were given the mark.

Many students were able to gain one mark in Q04.5 for stating that, as the nucleon numbers were the same, the value for the radius would be the same. Few were able to give any more information to support their reasoning, however.

The most common error in 04.6 was the use of the y-intercept as $\ln\,(R_{\mathrm{o}})$. Students who took this approach were limited to a maximum of two out of the four marks. Some students failed to get the first mark as their lines were not drawn carefully.

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QUESTION 05

In question 05.1, most students were able to obtain a reasonable estimate for the energy provided either by counting squares or making a reasonable judgement about the average power. Only those students who gave their answer as an order of magnitude were able to gain all three marks.

In Q05.2, power of ten errors were fairly common. Only about one-third of students received this mark either due to power of ten errors or for not using a reasonable value for the power. Students were expected to realise that the base-power stations produced the power that was provided at all times.

Most students were given one mark for Q05.3, usually for stating one similarity and one difference. Explanations were often too vague or incomplete and the second mark was rarely awarded.

The explanation required in Q05.4 also proved to be challenging for many students. The best answers made a clear comparison between the two types of power station and used the data to explain why more power was required from back-up power stations when less was available from solar.

QUESTION 06

The straightforward calculation in Q06.1 was answered correctly by most students. Those who failed to get both marks either did not use the mass of both spheres or, more rarely, made a rounding error.

Q06.2 proved to be even more straightforward with almost all the students who gained the first mark going on to get both marks.

The explanations in Q06.3 and Q06.4 were much more demanding. The best answers showed a good understanding of the context of the question and gave precise and logical responses. Answers to Q06.4 were often vague and incomplete and made little reference to the information given in the question.

QUESTION 07

Question 07.1 was not answered well by many students. Those who were given the mark demonstrated a good understanding of the difference between rest mass and the difference in masses between a nucleus and its nucleons. They then went on to relate these masses to their equivalent energies. Despite many approaches being available for credit, very students were given the mark.

In Q07.2, students who spotted that particle X was a neutron had little difficulty picking up both of the marks. The most straightforward solutions used the mass in u for the neutron and converted it into MeV. Using the mass in kg was also given credit unless the mass of the nucleon or the proton was used.

The calculation in Q07.3 required an understanding of binding energy in nuclear fusion. Only about one third of students were given both marks. Partial credit was available for adding the binding energy,

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provided it was sufficiently clear what the student was doing. This meant that more than half of the students were given at least one mark.

Many students were able to discuss one factor in sufficient detail to get one mark in Q07.4. Very few were given the second mark, however. Again, answers were either vague or incomplete.

SECTION B

The most accessible questions were B8, B9, B10, B12, B15 and B18. These were all answered correctly by more than three-quarters of the students.

The most demanding question was B20. This tested students' knowledge of critical mass. Both D and A were more popular than the correct answer.

Other questions with popular distractors were B19 and B14. In B19 many students chose B, using the height from the bottom of the reservoir perhaps. In B14 many chose A, which may have been due to some confusion about the roles of the moderator and the control rods.

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